# One-Dimensional Transient Heat Transfer with Source

## Question

Consider the data in the table, which describes a one-dimensional system subjected to a transient source. We shall determine the temperature as a function of time at five locations in the solid: $\xi = 0.0, 0.25, 0.5, 0.75,$ and 1.0. In obtaining the solution, we note that pdepe requires three functions: (1) one to define the partial differential equation, which we call pde1D; (2) one to define the initial conditions, which we call pdeIC; and (3) one to define the boundary conditions, which we call pdeBC. The boundary and initial conditions are summarized in the Table 1.

| Input Values | |
|---|---|
| Parameter | Value |
| Boundary Conditions and Source | |
| Dimensionless source strength: | $\Sigma = 1$ |
| At $\xi = 0$: | $Bi = 0.1$ |
| At $\xi = 1$: | $\theta(1, \tau) = \theta_w = 0.55$ |
| Initial Conditions | |
| Linear distribution: | $\theta(\xi, 0) = 1 - 0.45\xi$ |
| Numerical Grid Parameters | |
| Number of equally spaced grid points in | 41 |
| Number of equally spaced time steps | 101 |

**Table 1** Input Values

## Solution

### 1. Problem Statement

PDE:

$$\frac{\partial \theta}{\partial \tau} = \frac{\partial^2 \theta}{\partial \xi^2} + \Sigma, \qquad 0 < \xi < 1, \tau > 0,$$

with $\Sigma = 1$, $Bi = 0.1$, $T_r = 0.55$.

IC:

$$\theta(\xi, 0) = 1 - 0.45\xi$$

BCs:

$$\theta'(0, \tau) = Bi\theta(0, \tau), \qquad \theta(1, \tau) = T_r.$$

### 2. State State Solution

At steady state $\theta_s''(\xi) = -\Sigma = -1$. Integrating twice,

$$\theta_s(\xi) = -\frac{1}{2}\xi^2 + C_1\xi + C_2$$

BCs give.

$$-0.5 + C_1 + C_2 = 0.55 \;\rightarrow\; C_1 + C_2 = 1.05,$$
$$C_1 = BiC_2 = 0.1C_2$$

Solve exactly:

$$C_2 = \frac{1.05}{1.1} = \frac{21}{22}, C_1 = \frac{21}{220}$$

So, the exact steady state is.

$$\theta_S(\xi) = -\frac{\xi^2}{2} + \frac{21}{220}\xi + \frac{21}{22}$$

(checks: $\theta_s(1) = 0.55, \theta_s'(0) = 0.0954545 = 0.1 \cdot 0.954545.$)

## 3. Deviation and Homogeneous Problem

Define the deviation:

$$\phi(\xi,\tau) = \theta(\xi,\tau) - \theta_s(\xi).$$

Then $\phi$ satisfies the homogeneous heat equation.

$$\phi_\tau = \phi_{\xi\xi},$$

with homogeneous BCs,

$$\phi'(0,\tau) = Bi\phi(0,\tau), \qquad \phi(1,\tau) = 0,$$

and initial condition

$$\phi(\xi,0) = \phi_0(\xi) = 1 - 0.45\xi - \theta_s(\xi).$$

Compute $\phi_0(\xi)$ exactly (use rational coefficients):

$$\phi_0(\xi) = 1 - 0.45\xi + \frac{\xi^2}{2} - \frac{21}{220}\xi - \frac{21}{22}.$$

Combine terms (convert $0.45 = \frac{9}{20} = \frac{99}{220}$):

$$\text{coef of } \xi: \quad -\frac{99}{220} - \frac{21}{220} = -\frac{120}{220} = -\frac{6}{11},$$
$$\text{constant term:} \qquad 1 - \frac{21}{22} = \frac{1}{22}.$$

So,

$$\phi_0(\xi) = \frac{\xi^2}{2} - \frac{6}{11}\xi + \frac{1}{22}.$$

An especially useful simplification is that.

$$\phi_0 = (1-y) = \frac{1}{2}y^2 - \frac{5}{11}y,$$

which follows from substituting $y = 1 - \xi$ and simplifying (constant cancels).

## 4. Separation of Variables — eigenproblem

Assume $\phi(\xi,\tau) = X(\xi)T(\tau)$. We get $X'' + \lambda X = 0$ with $T(\tau) = e^{-\lambda\tau}$.

Set $\lambda = k^2$ with $k > 0$. General $X(\xi) = A\, cos(k\xi) + B \sin(k\xi)$. Using $X(1) = 0$ we can write eigenfunction as $\sin(k(1-\xi))$. The left BC $X'(0) = BiX(0)$ leads to the transcendental eigenvalue equation.

$$k \cot k = -Bi \qquad \text{(here } Bi = 0.1\text{)}.$$

This equation is transcendental: the roots $k_n$ (positive) must be found numerically. They lie once per interval $((n-1)\pi, n\pi)$.

Eigenfunctions: $X_n(\xi) = \sin(k_n(1-\xi))$ and decay factors $e^{-k_n^2\tau}$.

## 5. Exact Analytic Expression for the Coefficients $A_n$

We are expanding.

$$\phi(\xi,\tau) = \sum_{n=1}^{\infty} A_n e^{-k_n^2\tau} \sin(k_n(1-\xi)).$$

Coefficients by projection:

$$A_n = \frac{\int_0^1 \phi_0(\xi)\sin(k_n(1-\xi))d\xi}{\int_0^1 sin^2(k_n(1-\xi))d\xi}.$$

We evaluate numerators and denominators **analytically** in closed form (no unevaluated integrals left).

**Denominator (normalization)**

Using $y = 1-\xi$,

$$\int_0^1 sin^2\big(k(1-\xi)\big)d\xi = \int_0^1 sin^2(ky)dy = \frac{1}{2}\left(1 - \frac{\sin(2k)}{2k}\right).$$

**Numerator (explicit)**

Let $k = k_n$. Substitute $y = 1-\xi$. Using the simplification above,

$$I_n = \int_0^1 \phi_0(\xi)\sin\big(k(1-\xi)\big)\,d\xi = \int_0^1 \left(\frac{1}{2}y^2 - \frac{5}{11}y\right)\sin(ky)\,dy.$$

Compute the two elementary integrals analytically (integration by parts). Combining results gives the closed-form rational expression.

$$I_n = \frac{-k^2\, cos\, k + 12k\, sin\, k + 22\, cos - 22}{22\, k^3}.$$

**Final closed form for $A_n$**

Divide $I_n$ by the denominator $D_n = \frac{1}{2}\left(1 - \frac{\sin(2k)}{2k}\right)$ to obtain.

$$A_n = \frac{2(-k^2\cos k + 12k\, sin\, k + 22\cos k - 22)}{11\, k^2(2k - \sin(2k))}.$$

This is an exact algebraic expression in terms of $k = k_n$(no integrals left). All algebraic steps are standard integration by parts and rational simplification.

## 6. Eigenvalues $k_n$ – The Only Numerical Step

The only thing that cannot be given in elementary closed form is the sequence $\{k_n\}$: they are the positive roots of

$$k \cot k = -B_i \qquad \text{with } B_i = 0.1.$$

These roots must be found numerically (e.g., Newton or bisection); once you have $k_n$, the $A_n$ formula above is fully explicit.

For convenience, here are the first five eigenvalues and corresponding coefficients (computed numerically):

| n | $k_n$(approx.) | $A_n$(approx.) |
|---|---|---|
| 1 | 1.63199453 | -0.07329078 |
| 2 | 4.7335118 | -0.06724179 |
| 3 | 7.86669277 | 0.013591116 |
| 4 | 11.0046611 | -0.0105612 |
| 5 | 14.1442368 | 0.004784049 |

**Table 2** Numerical Step

## 7. Final Complete Solution

Putting everything together:

$$\theta(\xi,\tau) = \theta_s(\xi) + \sum_{n=1}^{\infty} A_n e^{-k_n^2\tau} \sin\left(k_n(1-\xi)\right)$$

With

$$\theta_s(\xi) = -\frac{\xi^2}{2} + \frac{21}{220}\xi + \frac{21}{22},$$

$k_n$ are the positive roots of $k \cot k = -Bi(Bi = 0.1)$,

$$A_n = \frac{2(-k_n^2 \cos k_n + 12k_n \sin k_n + 22\cos k_n - 22)}{11\, k_n^2(2k_n - \sin(2k_n))}.$$

That expression is fully explicit: the only practical step remaining is numeric root-finding for each $k_n$. One each $k_n$ is known, substituting it into the $A_n$ formula gives the coefficient exactly (no integrals to evaluate).

## 8. Practical Notes and Convergence

- The eigenvalues grow like $k_n \sim \left(n - \frac{1}{2}\right)\pi$ and eigenvalues in the exponent are $k_n^2$. Hence high modes decay extremely quickly with $\tau$. In practice, summing 4–8 terms give excellent accuracy for moderate $\tau$.
- Boundary checks:

- $\theta(1, \tau) = \theta_s(1) = 0.55$ for all $\tau$ because $\sin\big(k_n(1-1)\big) = 0$.
- As $\tau \to \infty$ the transient sum decays and $\theta \to \theta_s$.

## Result

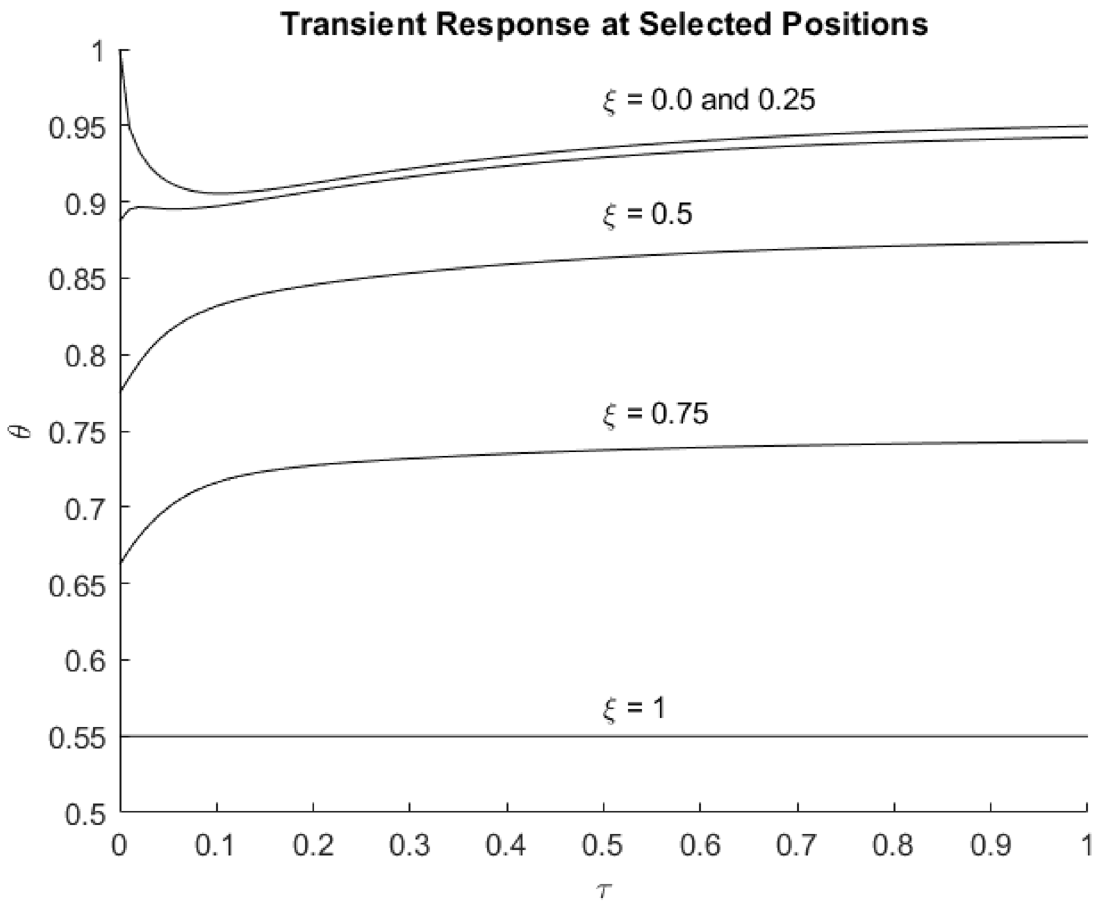


**Figure 1** One-dimensional heat conduction using the data in Table 1 Input Values.

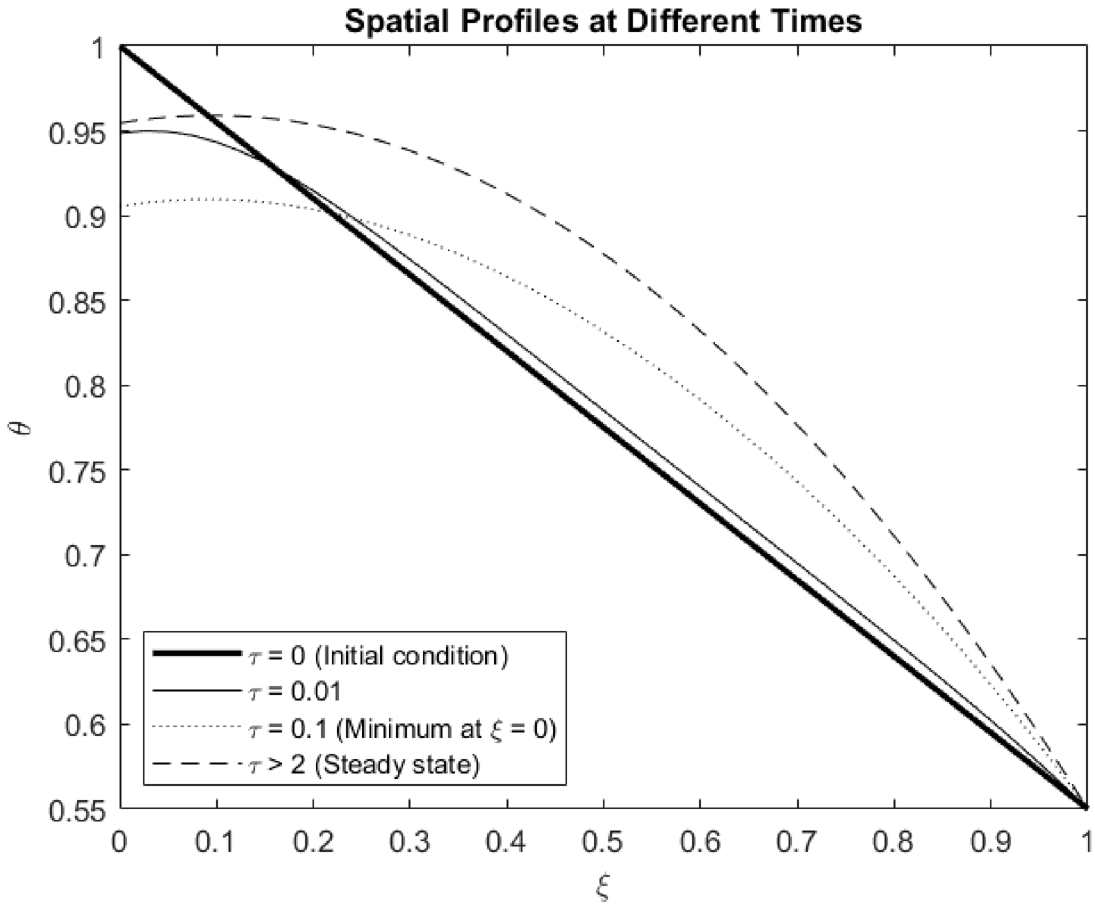


**Figure 2** One-dimensional heat conduction using the data in Table 1 Input Values.